# A guided tour in random forests

E. Scornet

Internal meeting - ANR GATSBII - November 2026






# Summary



1. Random Forests construction
2. Several random forest models
3. Number of trees
4. Centered forests
5. Mondrian Forests
6. Median forests
7. Breiman forests
8. Conclusion

# Statistical framework



Consider $(X_i, Y_i)$ i.i.d. copies of the pair $(X, Y)$

- **Input variable** $X \in \mathcal{X} = [0,1]^d$
- **Output variable** $Y \in \mathbb{R}$.
- **Regression model**: $Y = m(\mathbf{X}) + \varepsilon$.

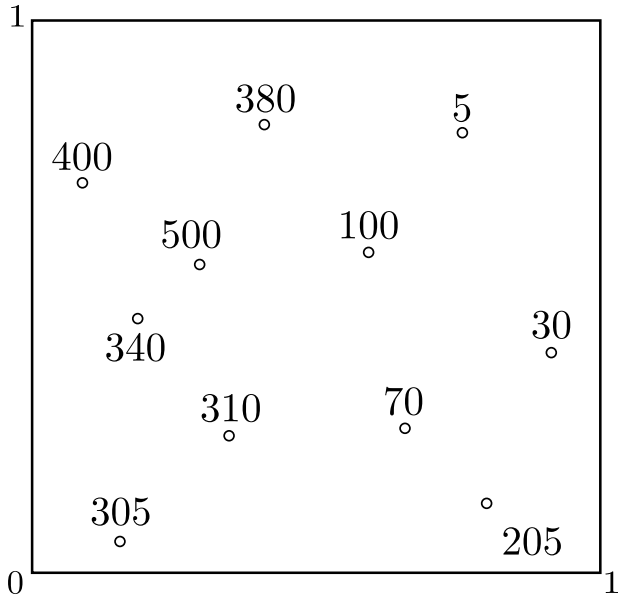
**Goal:** estimation of $m$ using **random forests**.



- Non-parametric method
- Based on bagging and random feature selections
- Aggregate the predictions of $M$ trees

# Construction of Decision trees - regression



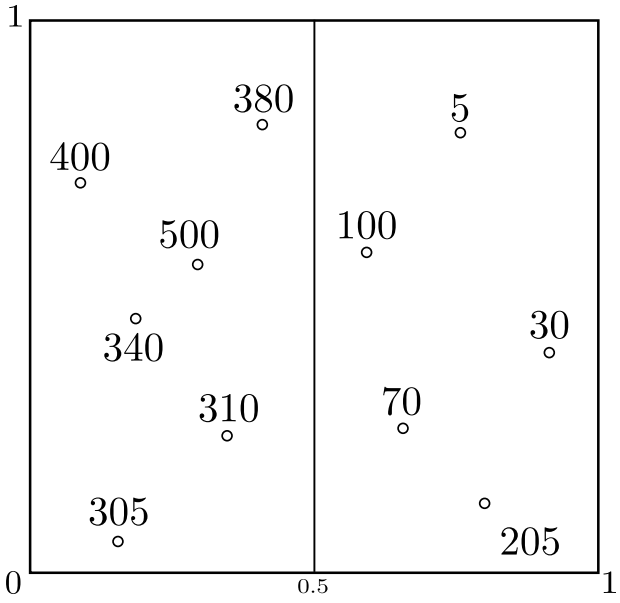



$k = 0$



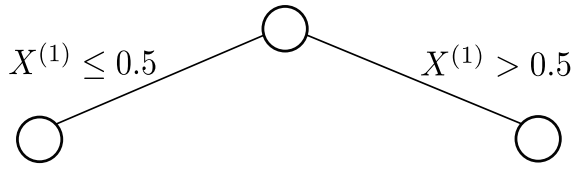
# Construction of Decision trees - regression





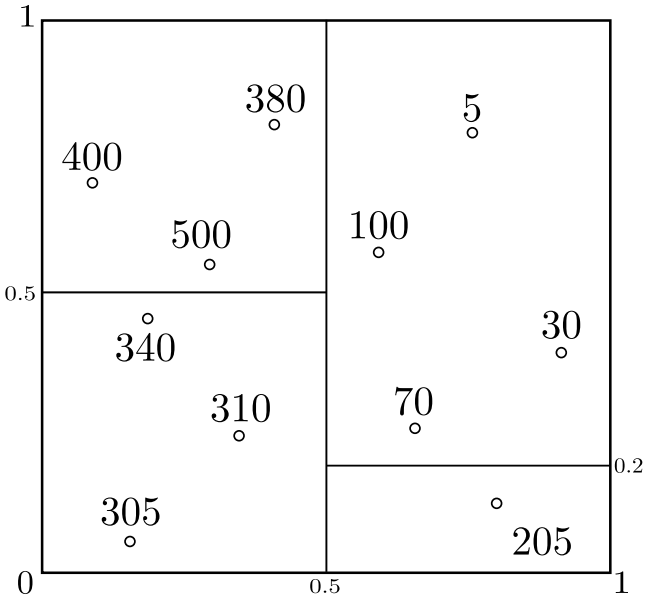

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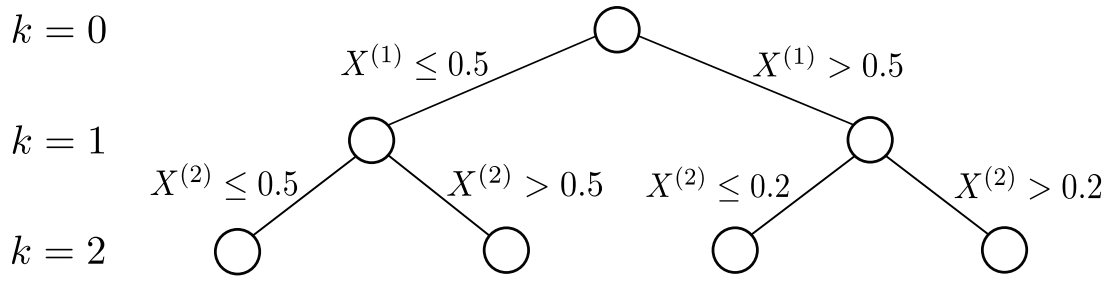
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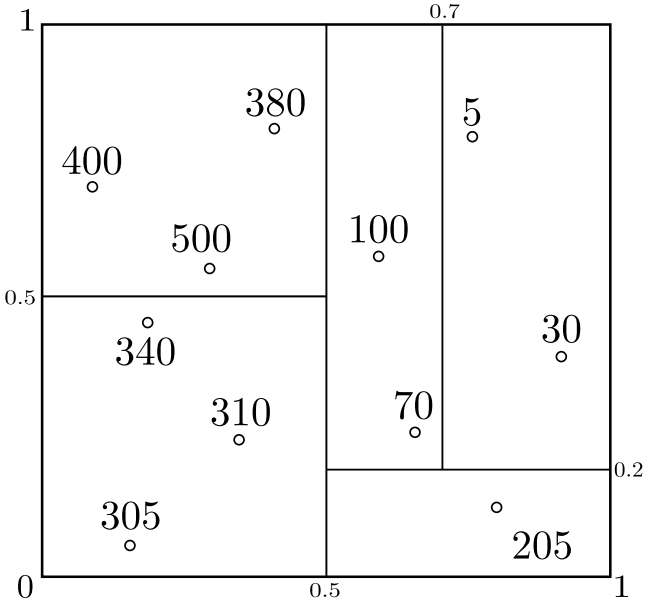




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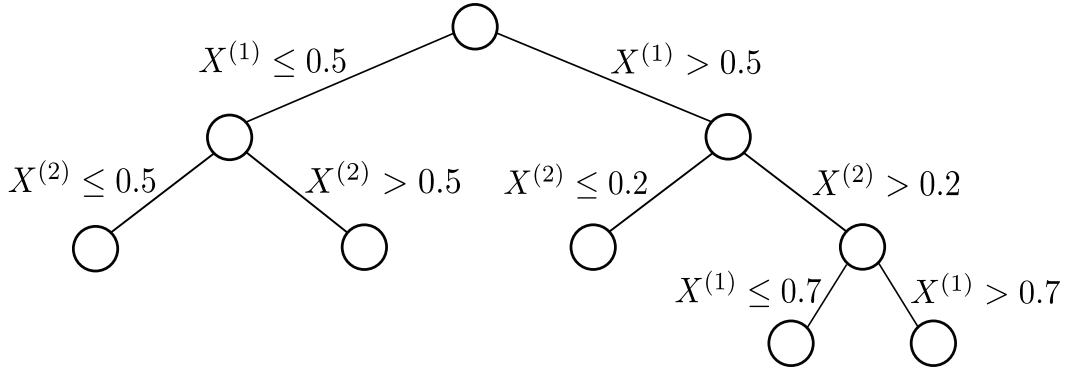



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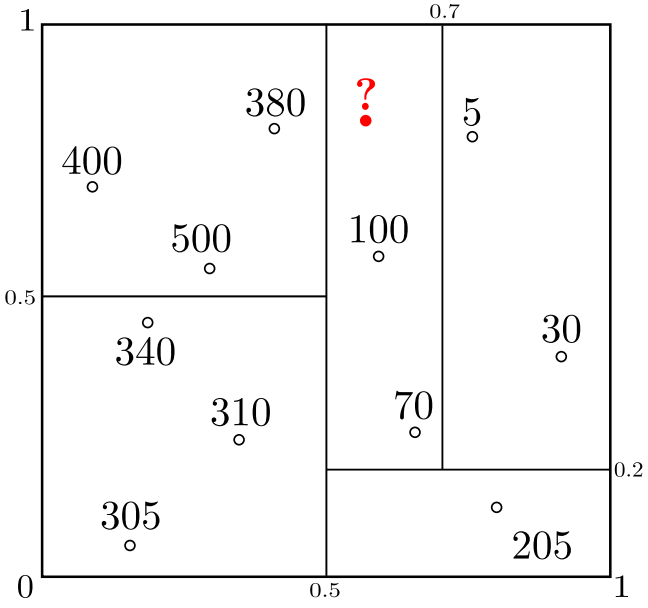
$k = 2$

$k = 3$



# Construction of Decision trees - regression



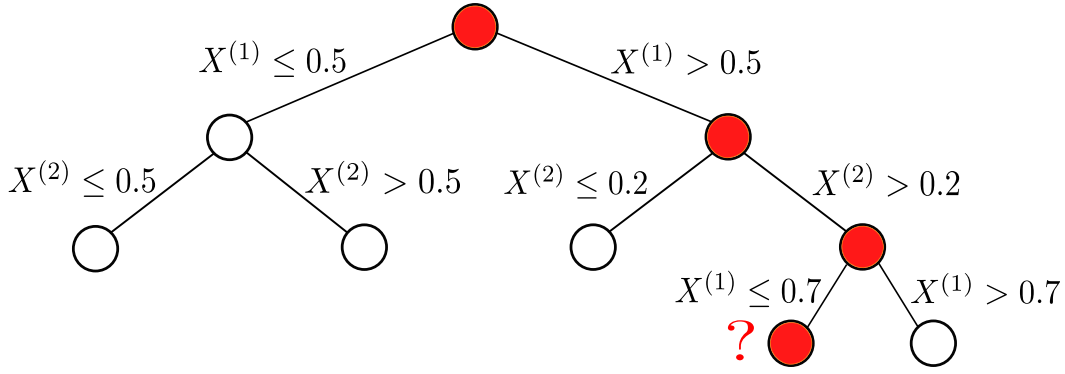



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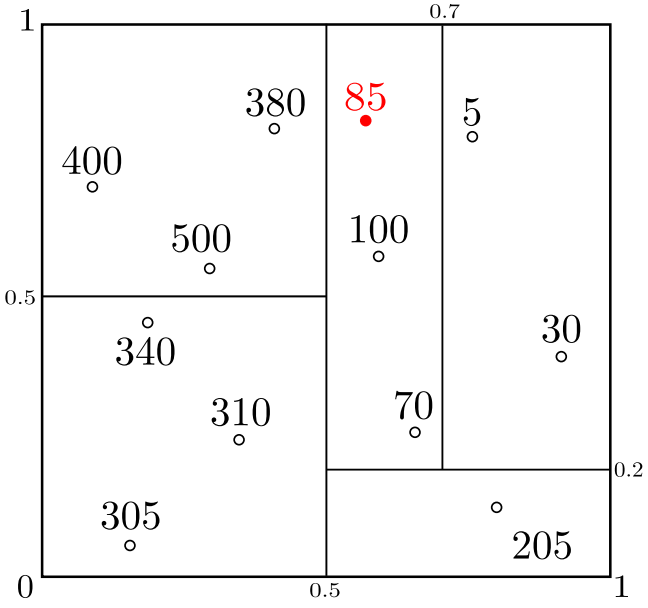
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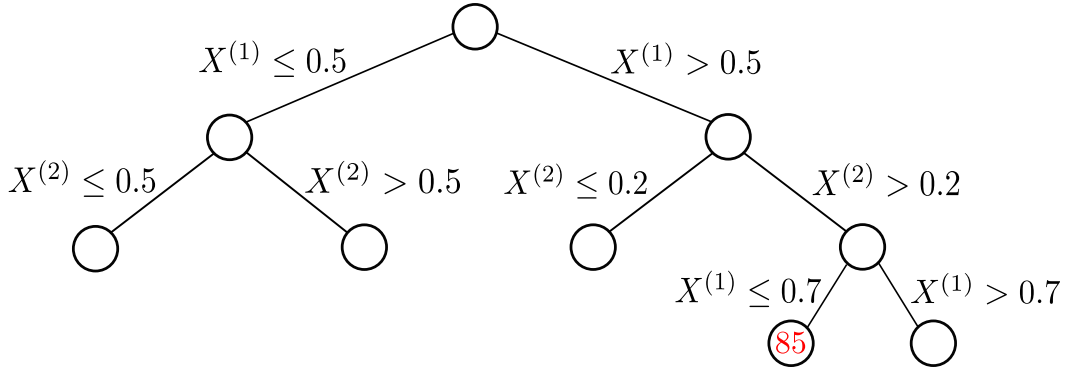



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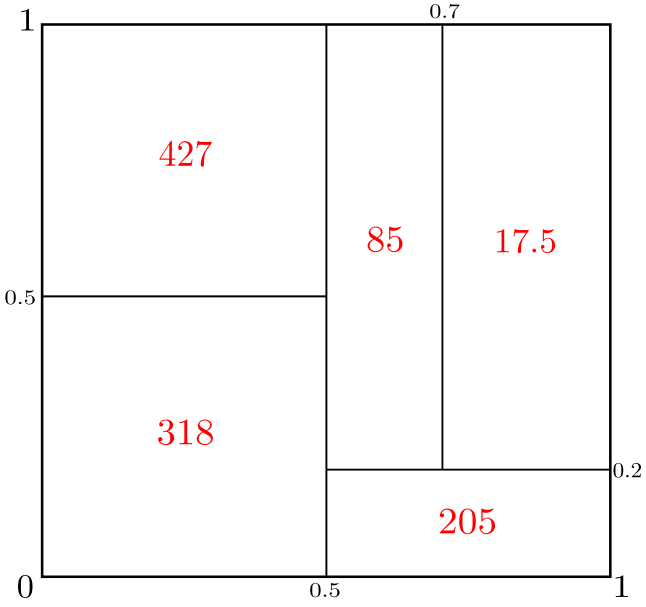
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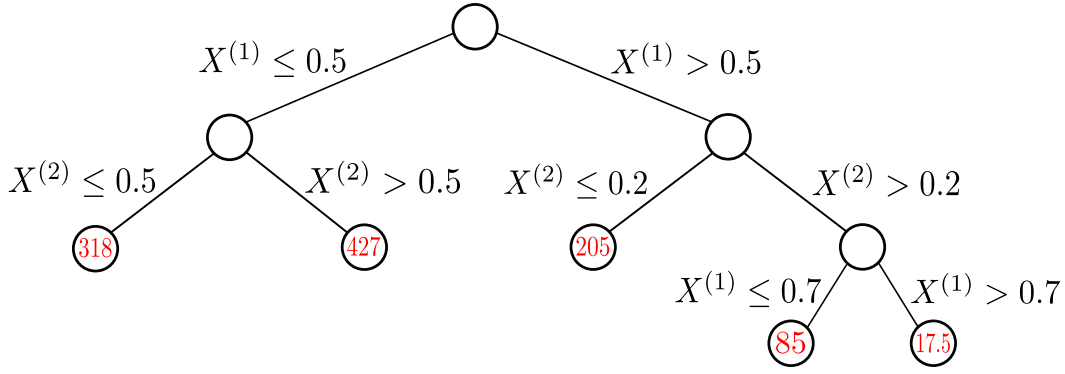



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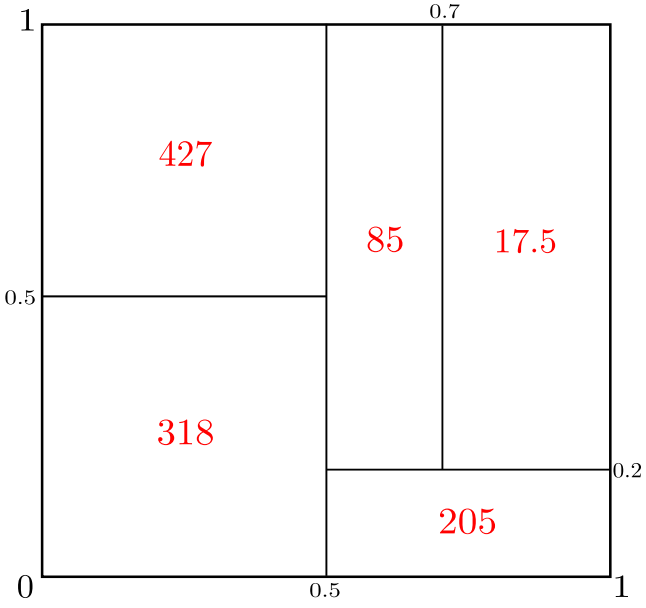
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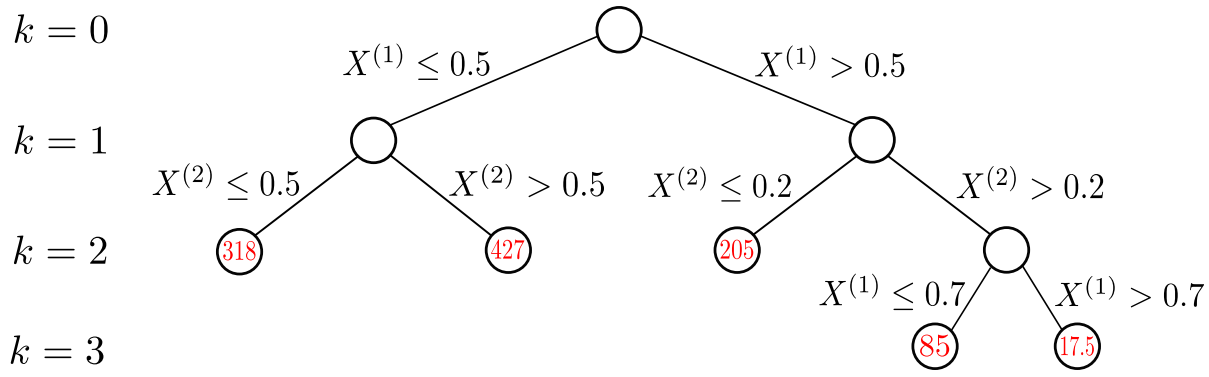
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# Construction of Decision trees - regression






Decrease in impurity for a split $(j, s)$:

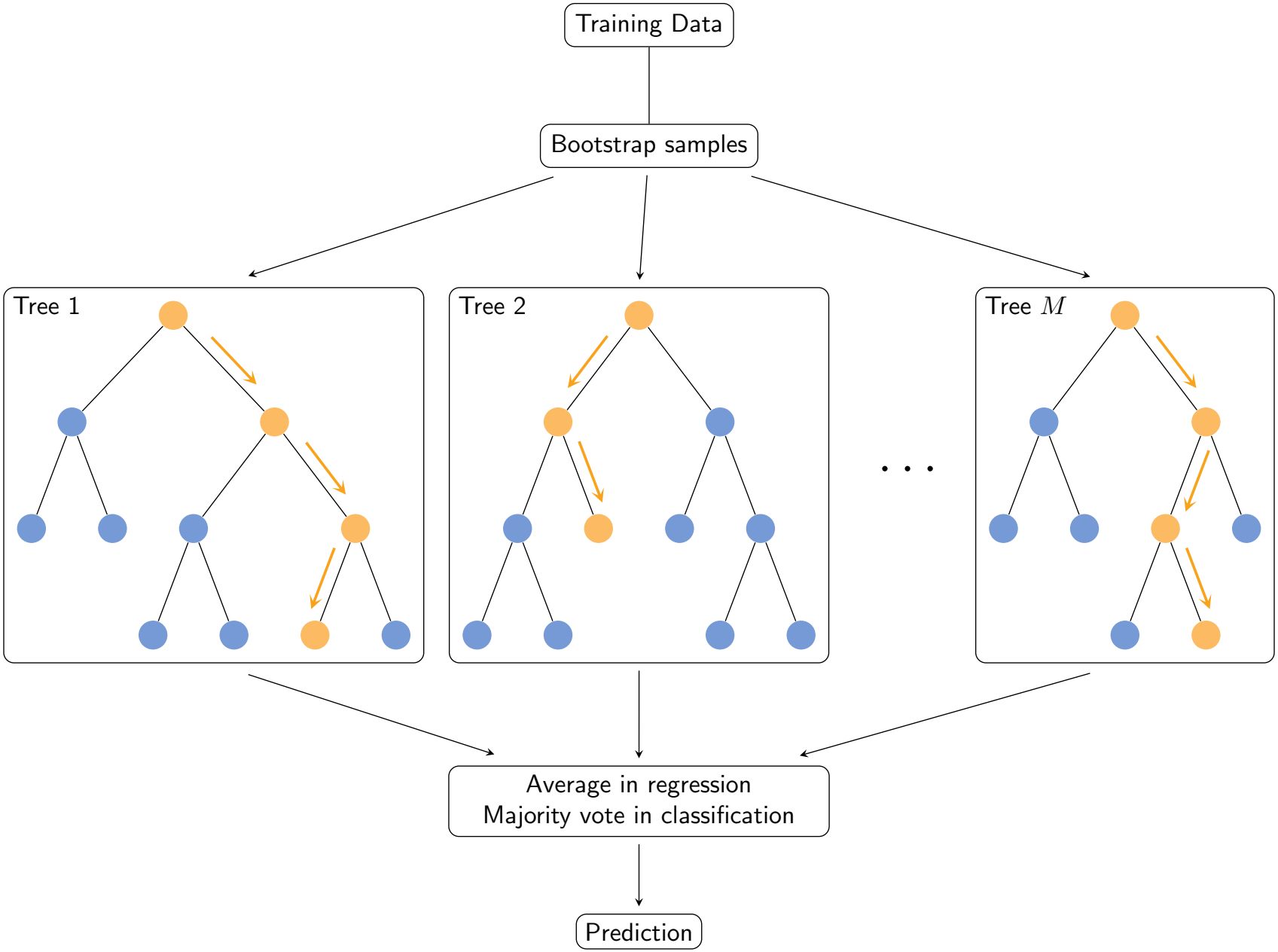
$$\Delta Imp(j, s; A) = Imp(A) - p_L Imp(A_L) - p_R Imp(A_R),$$

where $p_L$ (resp. $p_R$) is the fraction of observations in $A$ that fall into $A_L$ (resp. $A_R$). The best split $(j^\star, s^\star)$ is then chosen as

$$(j^\star, s^\star) \in \text{argmax}_{j,s} \Delta Imp(j, s; A).$$

# A classical random forest





# Theory of RF (Breiman, 2001): literature review



- **Simplified RF versions**, whose construction is **independent of the dataset**.

  (Biau et al., 2008; Biau, 2012; Genuer, 2012; Arlot and Genuer, 2014; Scornet, 2016b; Mourtada et al., 2020; Klusowski, 2021)

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- Analysis of more data-dependent forests:
  - **Asymptotic normality** of random forests (Mentch and Hooker, 2016; Wager and Athey, 2018),
  - **Variable importance** (Louppe et al., 2013; Li et al., 2019; Scornet, 2023),
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- Literature review on random forests:
  - **Methodological review** (Criminisi et al., 2011; Boulesteix et al., 2012),
  - **Theoretical review** (Biau and Scornet, 2016; Scornet and Hooker, 2026)

# Summary

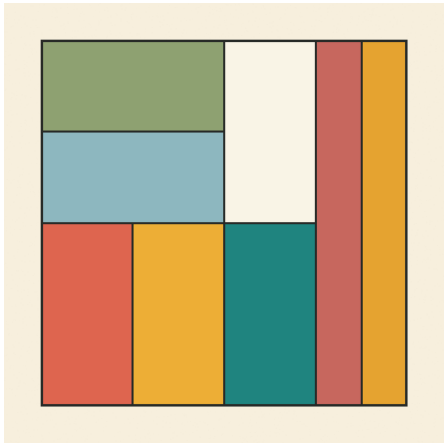


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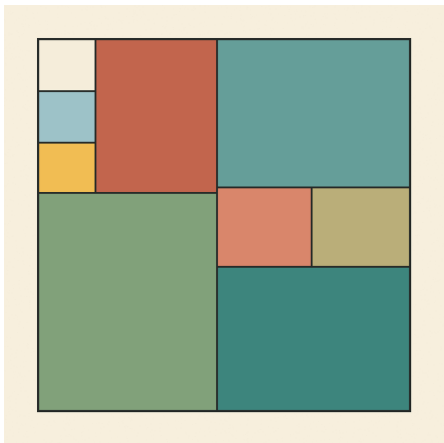
# Different types of forests



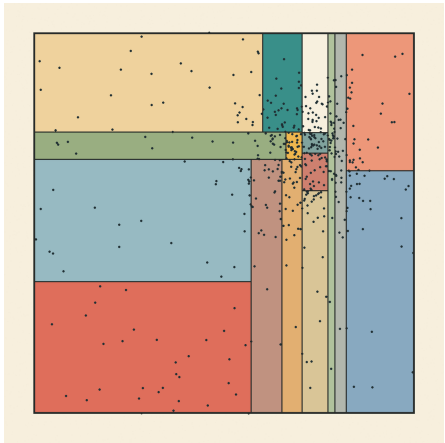
| Purely RF | Semi-adaptive RF | Adaptive RF |
|---|---|---|
| Independent of $X_i$ and $Y_i$ | Independent of $Y_i$ | Dependent on $X_i$ and $Y_i$ |



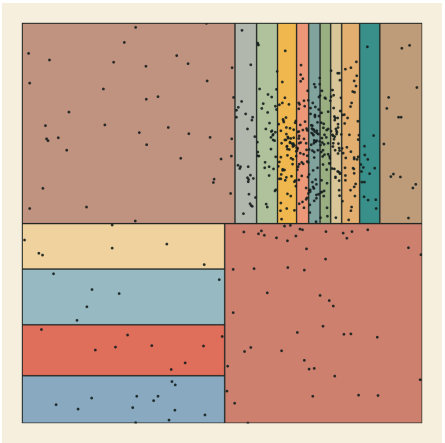
Centered RF



Mondrian RF



Median RF



Breiman RF

# Summary of random forests properties



| | Centred | Mondrian | Median | Breiman |
|---|---|---|---|---|
| **Data-dependent** | | | | |
| **Fully-developed** | | | | |
| **Consistency** | | | | |
| **Rate (Lipschitz)** | | | | |
| **Benefits of** | | | | |
| **Number of trees** | | | | |
| **Subsampling** | | | | |
| **Feature random.** | | | | |
| **Fully-developed** | | | | |

# Summary of random forests properties



| | Centred | Mondrian | Median | Breiman |
|---|---|---|---|---|
| **Data-dependent** | ✗ | ✗ | $\simeq$ | ✓ |
| **Fully-developed** | ✗ | ✗ | ✓ | ✓ |
| **Consistency** | | | | |
| **Rate (Lipschitz)** | | | | |
| **Benefits of** | | | | |
| **Number of trees** | | | | |
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| **Feature random.** | | | | |
| **Fully-developed** | | | | |

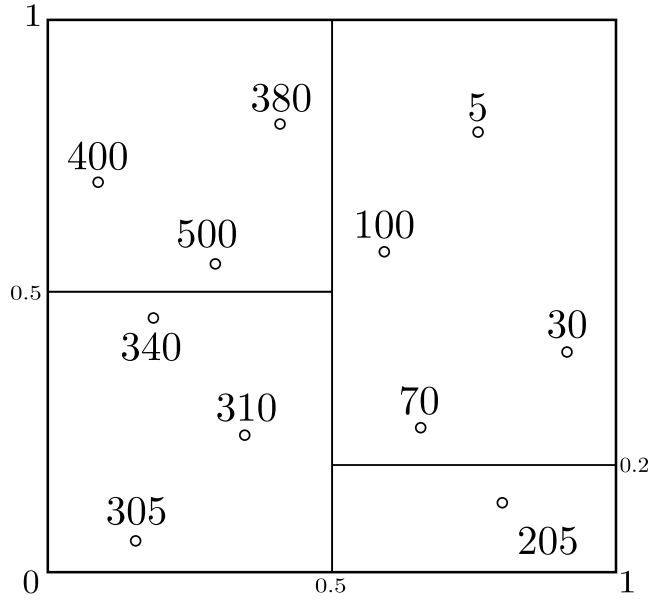
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# A tree





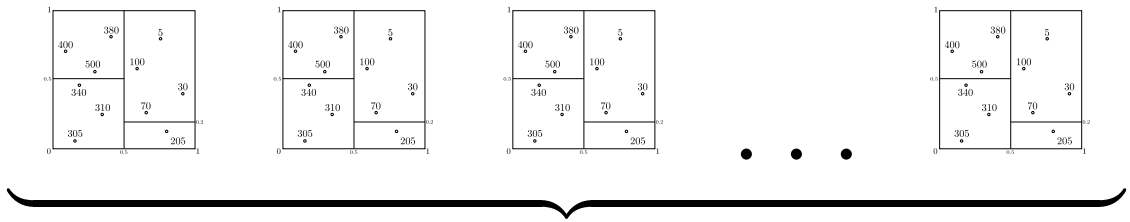

- Tree estimate:

$$m_n(\mathbf{x}, \Theta) = \sum_{i=1}^{n} \frac{\mathbb{1}_{\mathbf{X}_i \in A_n(\mathbf{x}, \Theta)}}{N_n(\mathbf{x}, \Theta)} Y_i$$

where $N_n(\mathbf{x}, \Theta)$ is the number of points in the cell $A_n(\mathbf{x}, \Theta)$.

# A finite forest



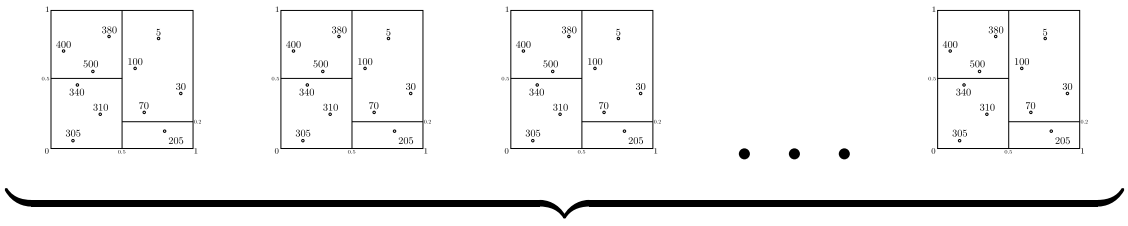



- $M$-Finite forest estimate :

$$m_{M,n}(\mathbf{x}, \Theta_1, \ldots, \Theta_M) = \frac{1}{M} \sum_{m=1}^{M} m_n(\mathbf{x}, \Theta_m).$$

# A finite forest





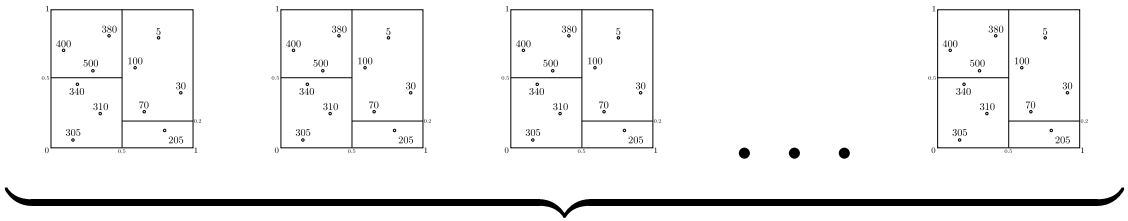

- $M$-Finite forest estimate :

$$m_{M,n}(\mathbf{x}, \Theta_1, \ldots, \Theta_M) = \frac{1}{M} \sum_{m=1}^{M} m_n(\mathbf{x}, \Theta_m).$$

Conditionally on $\mathcal{D}_n$, the estimate $m_{M,n}$ depends on $\Theta_1, \ldots, \Theta_M$.

# Toward infinite forest






- $M$-Finite forest estimate :

$$m_{M,n}(\mathbf{x}, \Theta_1, \ldots, \Theta_M) = \frac{1}{M} \sum_{m=1}^{M} m_n(\mathbf{x}, \Theta_m) \underset{M \to \infty}{\to} \underbrace{\mathbb{E}_\Theta \left[ m_n(\mathbf{x}, \Theta) \right]}_{m_{\infty,n}(\mathbf{x})}$$

# Finite forest versus infinite forest



Infinite forest is better than finite forest.

**(H1)** One has

$$Y = m(\mathbf{X}) + \varepsilon,$$

where $\varepsilon$ is a centered Gaussian noise with finite variance $\sigma^2$, independent of $\mathbf{X}$.

## Theorem (Scornet, 2016a)

Assume that **(H2)** is satisfied. Then, for all $M, n \in \mathbb{N}^\star$,

$$R(m_{M,n}) = R(m_{\infty,n}) + \frac{1}{M}\,\mathbb{E}_{\mathbf{X},\mathcal{D}_n}\Big[\mathbb{V}_\Theta\left[m_n(\mathbf{X},\Theta)\right]\Big].$$

In particular,

$$0 \leqslant R(m_{M,n}) - R(m_{\infty,n}) \leqslant \frac{8}{M} \times \left(\|m\|_\infty^2 + \sigma^2(1 + 4\log n)\right).$$

# A single tree versus a forest



**Theorem**

*We have*

$$\mathbb{E}[m_{M,n}(\mathbf{X}, \Theta_1, \ldots, \Theta_M) - m(\mathbf{X})]^2 \leqslant \mathbb{E}[m_n(\mathbf{X}, \Theta) - m(\mathbf{X})]^2,$$

*that is the risk of a forest is lower than the risk of each individual tree that composed the forest.*

**Proof.**

Jensen's inequality. □

A forest is not worse than a single tree.

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# A single tree





For a tree whose construction is independent of data, if

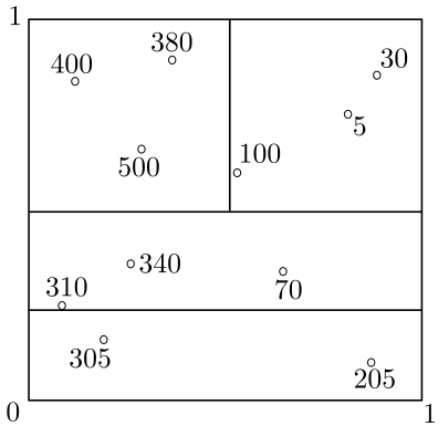
1. $\mathrm{diam}(A_n(\mathbf{X})) \to 0$, in probability;
2. $N_n(A_n(\mathbf{X})) \to \infty$, in probability;

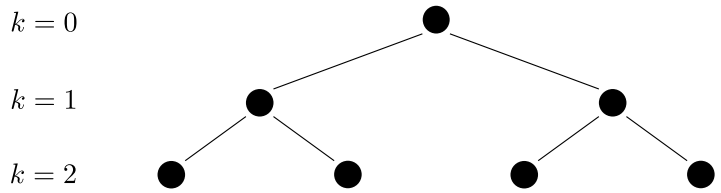
then the tree is consistent, that is

$$\lim_{n\to\infty} \mathbb{E}\left[m_n(\mathbf{X}) - m(\mathbf{X})\right]^2 = 0.$$

# Consistency of purely random forests






## Theorem (Biau et al., 2008)

Consider a totally non adaptive forest of level $k$. Assume that

$$\mathrm{diam}(A_n(\mathbf{X}, \Theta)) \to 0, \quad \text{in probability.}$$

Then, providing $k \to \infty$ and $n/2^k \to \infty$, the infinite random forest is consistent, that is $R(m_{\infty,n}) \to 0$ as $n \to \infty$.

- $\to$ Forest consistency results from the consistency of each tree.
- $\to$ Trees are not fully developed.

# Centered forests versus



Intermediate quantity $\bar{m}^{cc}_{\infty,n}(\mathbf{X}) = \sum_{i=1}^{n} W_{ni}(\mathbf{X}) m(\mathbf{X}_i)$.

## Estimation error (Biau, 2012)

$$\mathbb{E}\left[m^{cc}_{\infty,n}(\mathbf{X}) - \bar{m}^{cc}_{\infty,n}(\mathbf{X})\right]^2 \leqslant C\sigma^2 \frac{2^{k_n}}{n k_n^{1/2}}$$

## Approximation error (Biau, 2012)

$$\mathbb{E}\left[\bar{m}^{cc}_{\infty,n}(\mathbf{X}) - m(\mathbf{X})\right]^2 \leqslant 2dL^2.2^{-\frac{0.75k_n}{d\log 2}} + \|m\|^2_\infty e^{-n/2^{k_n}}$$

- Smaller variance due to averaging over more observations
- Randomness of tree partition harms the approximation error of the forest

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$$\leqslant L^2 2^{-2k_n/d} + \|m\|^2_\infty e^{-n/2^{k_n}}$$

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# Consistency of centred random forest



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$$\bar{m}^{cc}_{\infty,n}(\mathbf{X}) = \sum_{i=1}^{n} W_{ni}(\mathbf{X}) m(\mathbf{X}_i)$$

If the forest is fully grown, that is, if $k_n = \lfloor \log_2 n \rfloor$

## Estimation error (Biau, 2012)

Under proper assumptions on the regression model,

$$\mathbb{E}\left[m^{cc}_{\infty,n}(\mathbf{X}) - \bar{m}^{cc}_{\infty,n}(\mathbf{X})\right]^2 \leqslant C\sigma^2 (\log_2 n)^{-1/2}$$

## Approximation error (Biau, 2012)

Under proper assumptions on the regression model,

$$\mathbb{E}\left[\bar{m}^{cc}_{\infty,n}(\mathbf{X}) - m(\mathbf{X})\right]^2 \leqslant 2dL^2 2^{-\frac{0.75 k_n}{d \log 2}} + \|m\|^2_\infty e^{-n/2^{k_n}}$$

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## Risk of centered forests

Assume that $\mathbf{X}$ is uniformly distributed on $[0,1]^d$, $\|m\|_\infty < \infty$ and $\max_j \|\partial_j m\|_\infty < \infty$. Then,

$$\mathbb{E}\left[m^{cc}_{\infty,n}(\mathbf{X}) - m(\mathbf{X})\right]^2 \lesssim \sigma^2 \frac{2^{k_n}}{n k_n^{1/2}} + L^2 . 2^{-\frac{0.75 k_n}{d \log 2}} + \|m\|^2_\infty e^{-n/2^{k_n}}$$
$$\lesssim n^{-r},$$

with

- $r = \frac{1}{(4/3) d \log 2 + 1}$ (Breiman, 2004; Biau, 2012)

- Non minimax
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- $r = \frac{1}{(4/3) d \log 2 + 1}$ (Breiman, 2004; Biau, 2012)
- $r = \frac{1}{d \log 2 + 1}(1 + \delta)$ with $\delta \to 0$ as $d \to \infty$ (Klusowski, 2021)

- Non minimax
- Optimality of bias and variance for Lispchitz functions

# Summary

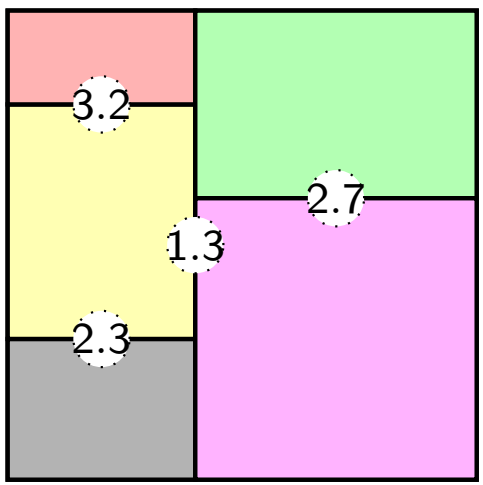


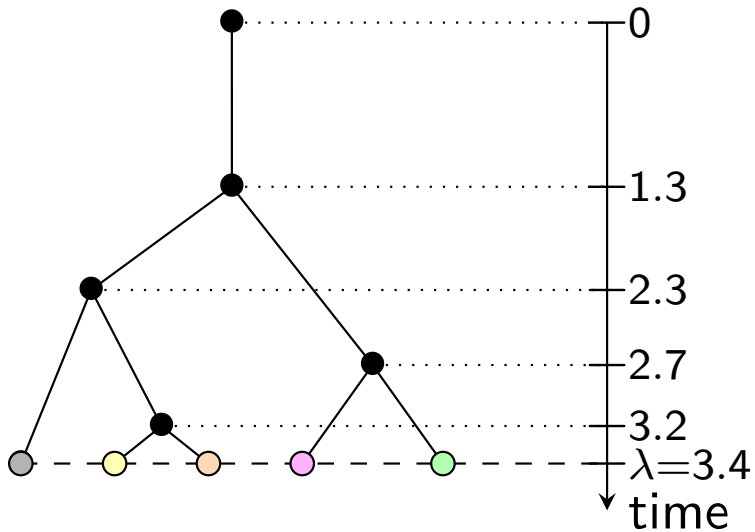
1. Random Forests construction
2. Several random forest models
3. Number of trees
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5. Mondrian Forests
6. Median forests
7. Breiman forests
8. Conclusion

# The Mondrian process (Roy and Teh, 2008)



- $\mathrm{MP}(\lambda, C)$: distribution on recursive, axis-aligned partitions of $C = \prod_{j=1}^{d}[a_j, b_j] \subset \mathbb{R}^d$ (= trees).
- $\lambda > 0$ "lifetime" = complexity parameter.

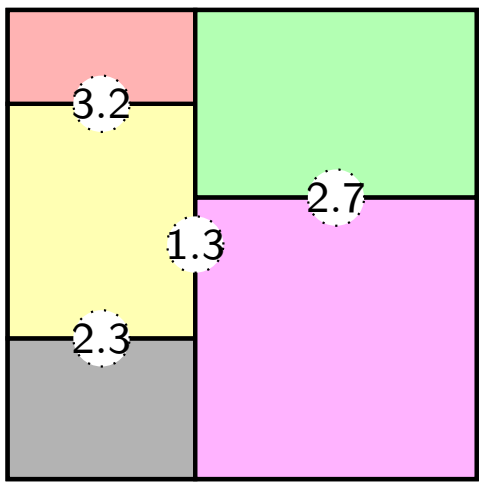


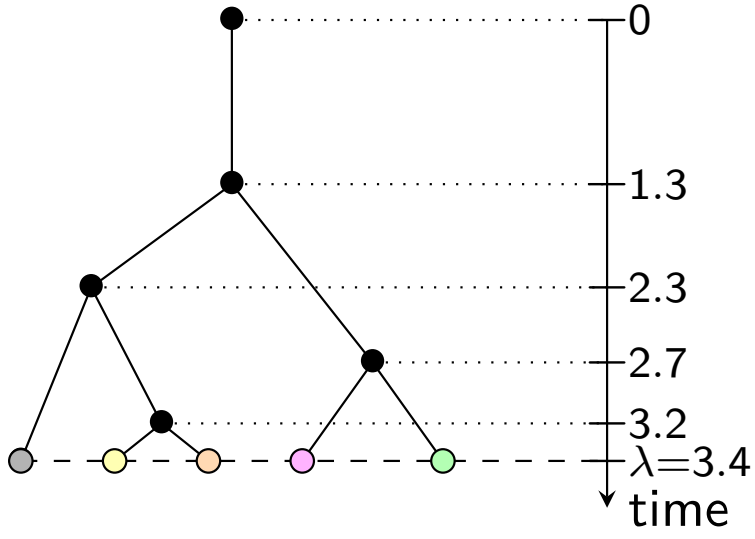



# The distribution $\mathrm{MP}(\lambda, C)$



- **Start** with cell $C$ (root), formed at time $\tau_C = 0$.
- **Sample** time till split $E \sim (|C|)$ with $|C| := \sum_{j=1}^{d}(b_j - a_j)$
- **If** $\tau_C + E \leqslant \lambda$,
  - split $C$ in $C_L = \{x \in C : x_J \leqslant S_J\}$ and $C_R = C \setminus C_L$:
    - split coordinate $J \in \{1, \ldots, d\}$ with $\mathbb{P}(J = j) = \frac{b_j - a_j}{|A|}$,
    - split threshold $S_J | J \sim ([a_J, b_J])$
  - Apply the procedure to $(C_L, \tau_C + E), (C_R, \tau_C + E)$.
- **Else** don't split $C$ (which becomes a leaf of the tree).



# Mondrian forests



- Introduced in Lakshminarayanan et al. (2014) for computational reasons: predictions updated efficiently with new sample point (online algorithm).
- Approximately: sample independent partitions $\Pi_\lambda^{(1)}, \ldots, \Pi_\lambda^{(M)} \sim \mathrm{MP}(\lambda, [0,1]^d)$, fit them and average their predictions.
- Purely random forests: construction independent of the dataset.
- No depth parameter but a lifetime parameter $\lambda$: tree depth increases with $\lambda$.

# Theoretical results



Denote $m_{\lambda,M,n}$ the (randomized) Mondrian forest estimator with $M$ trees and parameter $\lambda$.

## Theorem Mourtada et al. (2020)

Assume $\mathbb{V}(Y|X) \leqslant \sigma^2 < \infty$ and that the regression function $m$ is $L$-Lipschitz. Then:

$$\mathbb{E}[(Y - m_{\lambda,M,n}(\mathbf{X}))^2] \leqslant \frac{4dL^2}{\lambda^2} + \frac{(1+\lambda)^d}{n}\left(2\sigma^2 + 9\|m\|_\infty^2\right).$$

In particular, the choice $\lambda := \lambda_n \asymp n^{1/(d+2)}$ gives

$$R(m_{\lambda,M,n}) = O(n^{-2/(d+2)}),$$

which is the minimax optimal rate for the estimation of a Lipschitz function in dimension $d$.

# "Forest effect": influence of the number of trees



- The above result is true for every $M \geqslant 1$ (number of trees): in particular, a single tree is already optimal for the estimation of a Lipschitz function in dimension $d$.
- In practice, forests with $M \gg 1$ perform better than trees.
- How to account for this ? Do we gain something by randomizing partitions ?
- When is $M$ "large enough" ?

# Improved rates under $^2$ regularity



## Theorem Mourtada et al. (2020)

Assume **(H)**, $m$ of class $^2$, and that $\mathbf{X}$ has a positive, Lipschitz density on $[0,1]^d$. Then, for every $\varepsilon > 0$:

$$\mathbb{E}[(Y - m_{\lambda,M,n}(\mathbf{X})^2 \mathbb{1}_{[\varepsilon,1-\varepsilon]^d}] = O\Big(\frac{1}{M\lambda^2} + \frac{1}{\lambda^4} + \frac{e^{-\lambda\varepsilon}}{\lambda^3} + \frac{(1+\lambda)^d}{n}\Big)$$

For $\lambda := \lambda_n \asymp n^{1/(d+4)}$ and $M := M_n \gtrsim n^{2/(d+4)}$, this implies

$$\mathbb{E}[(Y - m_{\lambda,M,n}(\mathbf{X})^2 \mathbb{1}_{[\varepsilon,1-\varepsilon]^d}] = O(n^{-4/(d+4)})$$

which is the optimal rate for twice differentiable $m$ in dimension $d$. Without conditioning, we get $O(n^{-3/(d+3)})$ (boundary effect). By contrast, Mondrian trees do not exhibit improved rates.

Remark: Similar result obtained by Arlot and Genuer (2014) in dimension 1 for another variant of Random forests.

# Proof ideas



- Bias-variance decomposition: standard decomposition in approximation error + estimation error.
- Exact geometric properties (local and global) of Mondrian partitions are directly available, without reasoning conditionally on the graph structure / on earlier splits.
- Restriction property: enables to obtain the exact distribution of the cell $C_\lambda(x)$ of $x \in [0,1]^d$ in the partition $\Pi_\lambda \sim \mathrm{MP}(\lambda, [0,1]^d)$ (4 lines proof).
- By modifying the distribution of the Mondrian and using the one-dimensional case, one can show that the expected number of leaves in $\Pi_\lambda$ is $(1+\lambda)^d$.

# Summary



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# Construction of Median forests



## Median tree

- Select $a_n$ observations without replacement among the original sample $\mathcal{D}_n$. Use only these observations to build the tree.
- For each cell,
  - Select randomly $\texttt{mtry} = 1$ coordinate among $\{1, \ldots, d\}$;
  - Split at the location of the empirical median of $X_i$.
- Stop when each cell contains exactly $\texttt{nodesize} = 1$ observation.

# Consistency of median forests



## Assumption **(H1)**

The model writes $Y = m(\mathbf{X}) + \varepsilon$, where $\varepsilon$ is a centred noise such that $\mathbb{V}[\varepsilon|\mathbf{X} = \mathbf{x}] \leqslant \sigma^2$, $\mathbf{X}$ has a density on $[0,1]^d$ and $m$ is continuous.

## Theorem Scornet (2016a)

Grant Assumption **(H1)**. Then, provided $a_n \to \infty$ and $a_n/n \to 0$, median forests are consistent, i.e.,

$$\lim_{n\to\infty} \mathbb{E}\left[m_{\infty,n}(\mathbf{X}) - m(\mathbf{X})\right]^2 = 0.$$

- Good trade-off between simplicity of centred forests and complexity of Breiman's forests.
- First consistency results for fully grown trees.
- Each tree is not consistent but the forest is (subsampling).

# Proof of Theorem (1)



Condition $(i)$ in Stone's Theorem is satisfied since the regression function is uniformly continuous and $\mathrm{Var}[Y|\mathbf{X}=\mathbf{x}] \leqslant \sigma^2$ (see remark after Stone theorem in Györfi et al., 2002).

## Lemma 1

Assume that $\mathbf{X}$ has a density over $[0,1]^d$, with respect to the Lebesgue measure. Thus, the median tree satisfies, for all $\gamma$,

$$\mathbb{P}\left[\mathrm{diam}(A_n(\mathbf{X},\Theta)) > \gamma\right] \underset{n\to\infty}{\to} 0.$$

# Proof of Theorem (2)



To check (3), observe that in the subsampling step, there are exactly $\binom{n-1}{a_n-1}$ choices to pick a fixed observation $\mathbf{X}_i$. Since $\mathbf{x}$ and $\mathbf{X}_i$ belong to the same cell only if $\mathbf{X}_i$ is selected in the subsampling step, we see that

$$\mathbb{P}_\Theta\left[\mathbf{X} \overset{\Theta}{\leftrightarrow} \mathbf{X}_i\right] \leqslant \frac{\binom{n-1}{a_n-1}}{\binom{n}{a_n}} = \frac{a_n}{n}.$$

So,

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} W_{ni}(\mathbf{X})\right] \leqslant \mathbb{E}\left[\max_{1\leqslant i\leqslant n} \mathbb{P}_\Theta\left[\mathbf{X} \overset{\Theta}{\leftrightarrow} \mathbf{X}_i\right]\right] \leqslant \frac{a_n}{n},$$

which tends to zero by assumption.

# Construction of Median forests



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- Select $a_n$ observations without replacement among the original sample $\mathcal{D}_n$. Use only these observations to build the tree.
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# Rate of consistency



## Assumption **(H1)**

The regression model writes $Y = m(\mathbf{X}) + \varepsilon$, where $\mathbf{X}$ is uniformly distributed on $[0,1]^d$, $m$ is $L$-Lipschitz, and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.

## Theorem Duroux and Scornet (2018)

Grant **(H1)**. For all $n$, for all $\mathbf{x} \in [0,1]^d$, if $a_n \geqslant 2^k$, then

$$\mathbb{E}\big[m_{\infty,n}(\mathbf{x}) - m(\mathbf{x})\big]^2 \leqslant 2\sigma^2 \frac{2^k}{n} + dL^2 C_1 \left(1 - \frac{3}{4d}\right)^k.$$

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- Consistency result for each tree if $2^k / a_n \to 0$ and $k \to \infty$.

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- Right-hand side: Estimation error + approximation error.
- Consistency result for each tree if $2^k / a_n \to 0$ and $k \to \infty$.
- The upper bound is independent of $a_n$.

# Rate of consistency



Let $\beta = 1 - 3/(4d)$ and $k_n^\star = (\ln(n) + C_2)/(\ln 2 - \ln \beta)$.

## Theorem Duroux and Scornet (2018)

Assume that **(H1)** is satisfied. Consider a median forest of level $k = k_n^\star$. For all $n$, for all $\mathbf{x} \in [0,1]^d$, if $a_n \geqslant 2^{k_n^\star}$,

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The previous theorem holds in particular for two regimes:

- $a_n = n$: the median forest is not fully grown ($2^{k_n^\star}/n \to 0$) and the whole data set is used to build each tree.
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No need for tuning them both at the same time.

# Interpolation and non-parametric methods



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  No fixed number of parameters a priori

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$K(x) = \frac{1}{\|x\|^p}$

  - ✓ Interpolator
  - ✓ Consistent

[Devroye et al. 1998]
[Belkin et al. (2019)]

# Consistency of singular kernels



Belkin et al. (2019) consider Nadaraya-Watson predictors of the form

$$f_{a,h,n}(x) = \frac{\sum_{i=1}^{n} Y_i K_a\left(\frac{\|x-X_i\|}{h}\right)}{\sum_{i=1}^{n} K_a\left(\frac{\|x-X_i\|}{h}\right)},$$

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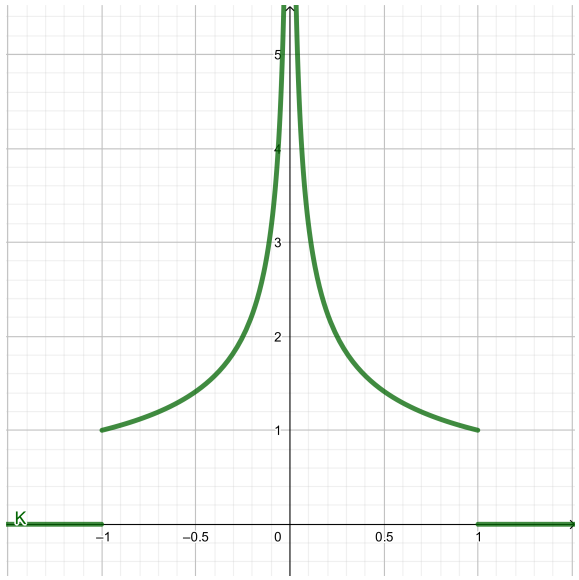



Figure: Singular kernel above for $a = 0.5$

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**Regression model:** $Y = f^\star(X) + \varepsilon$ with

- $\mathbb{E}[\varepsilon^2|X] \leqslant \sigma^2$ a.s.
- $X \sim \mathcal{U}([0,1]^d)$
- and $f^\star$ Lipschitz.

**Theorem (Belkin et al. (2019) - A specific case)**

*Let $0 < a < d/2$. Letting $h_n = n^{-1/(2+d)}$, we have*

$$Risk(f_{a,h_n,n}) \leqslant Cn^{-2/(d+2)}.$$

# Interpolation



- To analyze the strict interpolation case, we have to consider semi-adaptive RF

## Semi-adaptive median RF

1. Median tree
   - Select $a_n$ observations without replacement among the original sample $\mathcal{D}_n$. Use only these observations to build the tree.
   - For each cell,
     - Select randomly $\texttt{mtry} = 1$ coordinate among $\{1, \ldots, d\}$;
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2. Median RF: aggregation of median trees

# What we know about Median RF



## Assumption **(H1)**

The model writes $Y = m(X) + \varepsilon$, where $\varepsilon$ is a centred noise such that $\mathbb{V}[\varepsilon | X = x] \leqslant \sigma^2$, $X$ has a density on $[0,1]^d$ and $f^\star$ is continuous.

## Theorem (Scornet, 2016a)

Grant Assumption **(H1)**. Then, provided $a_n \to \infty$ and $a_n/n \to 0$, the infinite median forest is consistent, i.e.,

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**Remarks**

- Unsatisfying result because forest interpolation only occurs when $a_n = n$.

# Semi-adaptive RF: median RF



## Theorem (Arnould et al., 2023)

Suppose that $m$ has bounded partial derivatives and that $n$ is a power of two. Then, the interpolating Median RF $^{\mathrm{MedRF}}$ verifies:

$$\mathbb{E}\big[m_{\infty,n}(\mathbf{x}) - m(\mathbf{x})\big]^2 \leqslant C_1 d \left(\sum_{\ell=1}^{d} ||\partial_\ell f^\star||_\infty^2\right) \left(1 - \frac{3}{4d}\right)^{\log_2 n} + \sigma^2 C_{2,d} (\log_2 n)^{-(d-1)/2},$$

where $C_1$ and $C_{2,d}$ are explicit constants.

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## Theorem (Arnould et al., 2023)

Suppose that $m$ has bounded partial derivatives and that $n$ is a power of two. Then, the interpolating Median RF $^{\mathrm{MedRF}}$ verifies:

$$\mathbb{E}\big[m_{\infty,n}(\mathbf{x}) - m(\mathbf{x})\big]^2 \leqslant C_1 d \left(\sum_{\ell=1}^{d} ||\partial_\ell f^\star||_\infty^2\right) \left(1 - \frac{3}{4d}\right)^{\log_2 n} + \sigma^2 C_{2,d} (\log_2 n)^{-(d-1)/2},$$

where $C_1$ and $C_{2,d}$ are explicit constants.

- Interpolating (median) RF are consistent in a noisy setting (first result).
- Each tree is not consistent but the forest is (due to the randomization of splitting directions).
- First result to highlight the asymptotic benefit of split randomization (making the forest consistent).

# Unreachable minimax rate for purely random forests



Let $Y = m(\mathbf{X}) + \varepsilon$ with $\mathbb{E}(\varepsilon) = 0$, $\text{var}(\varepsilon) = \sigma^2$ and $\mathbf{X} \sim \mathcal{U}([0,1]^d)$.

## Theorem Lin and Jeon (2006)

Consider an estimator $\hat{m}_{\ell,n}$ resulting from a nonadaptive random forest with terminal node size at most $\ell$. Then, there exists $\Lambda > 0$ such that for any $n$,

$$\mathbb{E}[(Y - \hat{m}_{\ell,n}(\mathbf{X}))^2] \geqslant \Lambda \frac{(\log n)^{-(d-1)}}{\ell}.$$

# Summary



1. Random Forests construction
2. Several random forest models
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# Construction of Breiman forests



## Our Breiman tree

- Select $a_n$ observations without replacement among the original sample $\mathcal{D}_n$. Use only these observations to build the tree.
- For each cell,
  - Select randomly `mtry` coordinates among $\{1, \ldots, d\}$;
  - Choose the best split along previous direction, the one minimizing the CART criterion.
- Stop when the number of cells is exactly $t_n$.

# Additive regression model



## Assumption **(H1)**

The regression model is

$$Y = \sum_{j=1}^{d} m_j(\mathbf{X}^{(j)}) + \varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ with $\varepsilon$ independent of $\mathbf{X}$; $\mathbf{X}$ is uniformly distributed on $[0,1]^d$; each model component $m_j$ is continuous.

# Consistency



## Assumption **(H1)**

The regression model is $Y = \sum_{j=1}^{d} m_j(\mathbf{X}^{(j)}) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ with $\varepsilon$ independent of $\mathbf{X}$; $\mathbf{X}$ is uniformly distributed on $[0,1]^d$; each model component $m_j$ is continuous.

## Theorem (Scornet et al., 2015)

Assume that **(H1)** is satisfied. Then, provided $a_n \to \infty$ and $t_n(\log a_n)^9/a_n \to 0$, random forests are consistent, i.e.,

$$\lim_{n\to\infty} \mathbb{E}\left[m_{\infty,n}(\mathbf{X}) - m(\mathbf{X})\right]^2 = 0.$$

**Remarks**

- First consistency result for Breiman's original forest.
- Consistency of CART.

# Oracle inequality for CART



Let $\mathcal{G}^1 = \{g(\mathbf{x}) = g_1(x_1) + \ldots, +g_d(x_d)\}$, where $g_j : \mathbb{R} \to \mathbb{R}$ are measurable, with

$$\|g\|_{TV} = \sum_{j=1}^{d} TV(g_j), \quad \text{and} \quad TV(g_j) = \int |g_j'(u)| du.$$

## Theorem - Klusowski and Tian (2024)

Assume that the noise is sub-Gaussian, then $\forall K \geqslant 1$,

$$0.5\mathbb{E}\Big[(\widehat{m}_K^{CART}(\mathbf{X}) - m(\mathbf{X}))^2\Big] \leqslant \inf_{g(\cdot)\in\mathcal{G}^1} \Big\{\mathbb{E}[(g(\mathbf{X}) - m(\mathbf{X}))^2] + \frac{\|g\|_{\mathrm{TV}}^2}{K+3} + C_1 \frac{2^K \log^2(N)\log(Nd)}{N}\Big\}$$

where $C_1 > 0$ depends on $\|m\|_\infty$ and $\sigma^2$ (sub-Gaussian variance).

# Oracle inequality for CART



## Theorem - Klusowski and Tian (2024)

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where $C_1 > 0$ depends on $\|m\|_\infty$ and $\sigma^2$ (sub-Gaussian variance).

- Consistency of generic decision trees for additive functions.
- Upper bound on the rate of consistency for CART $(\log n)^{-1}$.

# Oracle inequality for CART



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where $C_1 > 0$ depends on $\|m\|_\infty$ and $\sigma^2$ (sub-Gaussian variance).

- Consistency of generic decision trees for additive functions.
- Upper bound on the rate of consistency for CART $(\log n)^{-1}$.
- Theorem 7.1: split randomization (mtry) cannot damage too much the predictive performances of RF.

# Oracle inequality for CART



## Theorem - Klusowski and Tian (2024)

Assume that the noise is sub-Gaussian, then $\forall K \geqslant 1$,

$$0.5\mathbb{E}\left[(\widehat{m}_K^{CART}(\mathbf{X}) - m(\mathbf{X}))^2\right] \leqslant \inf_{g(\cdot)\in\mathcal{G}^1} \left\{\mathbb{E}[(g(\mathbf{X}) - m(\mathbf{X}))^2] + \frac{\|g\|_{\mathrm{TV}}^2}{K+3} + C_1 \frac{2^K \log^2(N)\log(Nd)}{N}\right\}$$

where $C_1 > 0$ depends on $\|m\|_\infty$ and $\sigma^2$ (sub-Gaussian variance).

- Consistency of generic decision trees for additive functions.
- Upper bound on the rate of consistency for CART $(\log n)^{-1}$.
- Corollary 7.2: consistency of RF in sparse high-dimensional settings, where the dimension d is allowed to grow subexponentially with n.

# Lower bound for additive models



- Input $\mathbf{X} \in [0,1]^d$
- Additive model $m(\mathbf{x}) = \sum_{j=1}^{d} m_j(x_j)$
- Noise $\sigma^2 = \mathbb{E}[\varepsilon^2 | \mathbf{X} = \mathbf{x}]$

Consider an honest tree $\hat{m}(\cdot, \mathfrak{p})$ where $\mathfrak{p}$ is the admissible partition:

$$\mathcal{A} = \{\text{data-independent partitions } \mathfrak{p} \ s.t. \forall C \in \mathfrak{p}, \mathbb{P}(\mathbf{X} \in C) \geqslant 1/n\}$$

## Theorem 5.1 Tan et al. (2022)

Assume that for all $j = 1, \ldots, d$, $m_j \in C^1([0,1])$ and

$$\inf_{t \in [0,1]} |m_j'(t)| \geqslant \beta_0 > 0 \qquad \text{for all } j \in S.$$

Assume that $X$ has a continuous density $q$ such that $q \geqslant q_{min} > 0$. Then

$$\mathbb{E}\left[(\hat{m}(\mathbf{X}, \mathfrak{p}) - m(\mathbf{X}))^2\right] \geqslant d \left(\frac{\beta_0^2 q_{\min}}{12}\right)^{\frac{d}{d+2}} \left(\frac{\sigma^2}{4n}\right)^{\frac{2}{d+2}}.$$

Vitesse minimax pour un modèle additif avec fonctions $C^1$ : $n^{-2/3}$.

# Breiman RF and interpolation

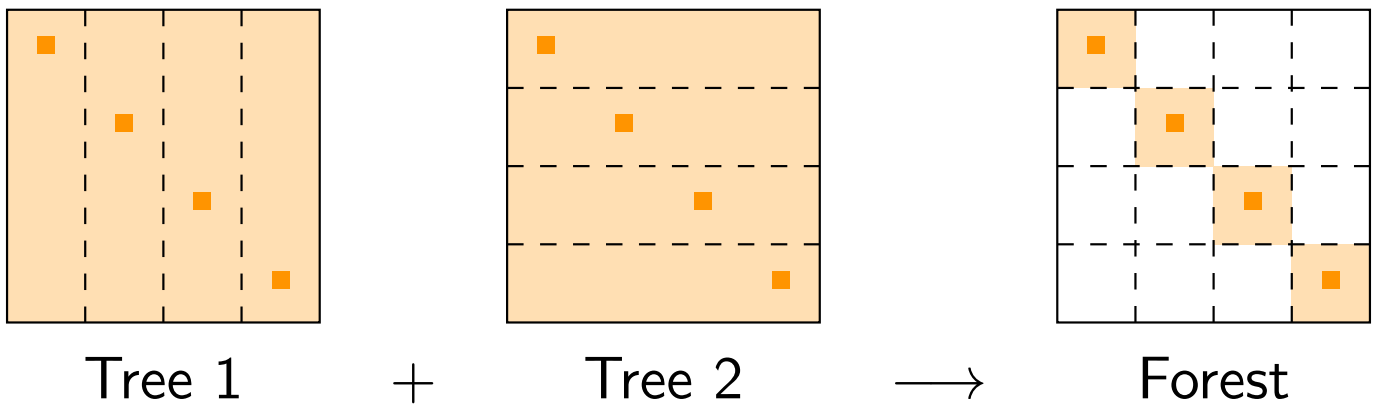


- ✗ Theoretical analysis of interpolating Breiman RF consistency: out of reach for now
- ▶ Study of the interpolation zone instead!
- ▶ Partition of the RF $\equiv$ intersection of the partitions of the trees of the RF

# Breiman RF and interpolation



- ✗ Theoretical analysis of interpolating Breiman RF consistency: out of reach for now
- ▶ Study of the interpolation zone instead!
- ▶ Partition of the RF $\equiv$ intersection of the partitions of the trees of the RF




**Interpolation zone**

Area of the space where the prediction relies on only one point of the dataset

# Breiman RF: volume of the interpolation zone



## Proposition Arnould et al. (2023)

Consider an infinite Breiman forest constructed without bootstrap, with `mtry` fixed to 1. Then, the volume of its interpolation zone $Z_n$ verifies

$$\mathbb{E}\left[\text{vol}(Z_n)\right] \leqslant \frac{1}{n^{d-1}}(1-2^{-n})^d$$

- The risk can be decomposed as

$$\begin{aligned}
&\text{Risk}(f_n(X)) - \text{Risk}(f^\star) \\
&= \underbrace{\text{Risk}((f_n(X) - f^\star(X))\mathbb{1}_{X\in Z_n})}_{\geqslant \sigma^2 \text{vol}(Z_n)} + \text{Risk}((f_n(X) - f^\star(X))\mathbb{1}_{X\notin Z_n})
\end{aligned}$$

- Necessary condition for consistency: $\text{vol}(Z_n) \to 0$ as $n \to \infty$
- For most points of the space, more than one point are involved in the prediction of the RF.

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# Summary of random forests properties

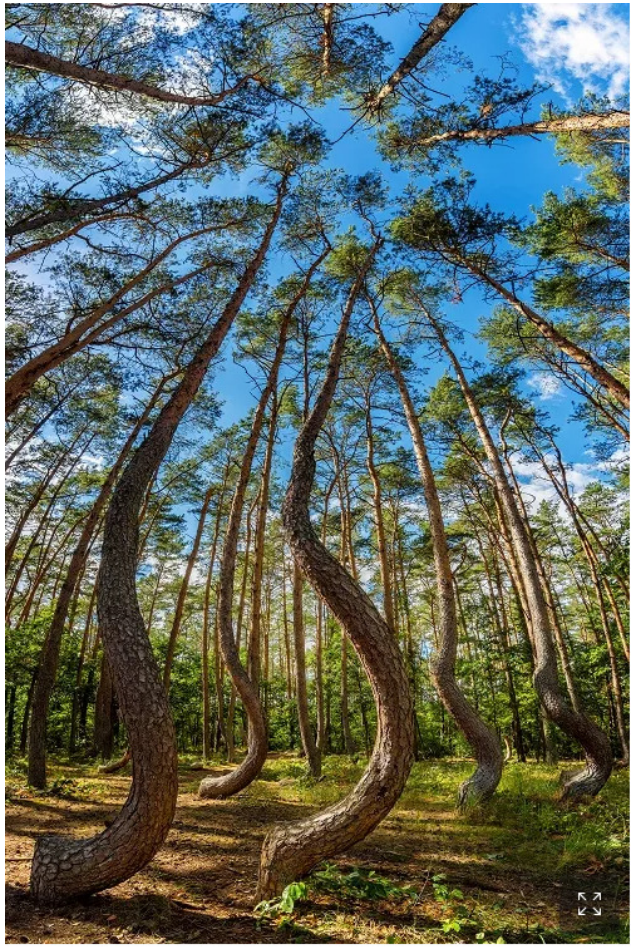


| | **Centred** | **Mondrian** | **Median** | **Breiman** |
|---|---|---|---|---|
| **Data-dependent** | ✗ | ✗ | $\simeq$ | ✓ |
| **Fully-developed** | ✗ | ✗ | ✓ | ✓ |
| **Consistency** | ✓ | ✓ | ✓ | ✓[1] |
| **Rate (Lipschitz)** | $n^{-1/(1+d\log 2)}$ | $n^{-2/(d+2)}$ | $(\log n)^{-(d-1)/2}$ | $(\log n)^{-1}$ (meas. funct.) |
| **Benefits of** | | | | |
| **Number of trees** | Var. reduct. | $C^2$-minimax | Consistency | ✗ |
| **Subsampling** | NA | NA | ✓ | ✗ |
| **Feature random.** | ✓ | ✓ | ✓ | ✗ |
| **Fully-developed** | NA | NA | ✗ | ✗ |

[1] Additive models

# Thank you for your attention!





Asymptotic Normality of Infinite Centered Random Forests - Application to Imbalanced Classification, M. Mayala, E. Scornet, C. Tillier and O. Wintenberger

# References I



S. Arlot and R. Genuer. Analysis of purely random forests bias. *arXiv preprint arXiv:1407.3939*, 2014.

Ludovic Arnould, Claire Boyer, and Erwan Scornet. Is interpolation benign for random forest regression? In *International Conference on Artificial Intelligence and Statistics*, pages 5493–5548. PMLR, 2023.

Mikhail Belkin, Alexander Rakhlin, and Alexandre B Tsybakov. Does data interpolation contradict statistical optimality? In *The 22nd international conference on artificial intelligence and statistics*, pages 1611–1619. PMLR, 2019.

G. Biau. Analysis of a random forests model. *JMLR*, 2012.

G. Biau and E. Scornet. A random forest guided tour. *Test*, 25:197–227, 2016.

G. Biau, L. Devroye, and G. Lugosi. Consistency of random forests and other averaging classifiers. *JMLR*, 9(9), 2008.

A.-L. Boulesteix, S. Janitza, J. Kruppa, and I.R. König. Overview of random forest methodology and practical guidance with emphasis on computational biology and bioinformatics. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 2:493–507, 2012.

L. Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.

# References II



L. Breiman. *Consistency for a simple model of random forests*. Technical Report 670, UC Berkeley, 2004.

A. Criminisi, J. Shotton, and E. Konukoglu. Decision forests: A unified framework for classification, regression, density estimation, manifold learning and semi-supervised learning. *Foundations and Trends in Computer Graphics and Vision*, 7:81–227, 2011.

Roxane Duroux and Erwan Scornet. Impact of subsampling and tree depth on random forests. *ESAIM: Probability and Statistics*, 22:96–128, 2018.

R. Genuer. Variance reduction in purely random forests. *Journal of Nonparametric Statistics*, 24(3):543–562, 2012.

L. Györfi, M. Kohler, A. Krzyżak, and H. Walk. *A Distribution-Free Theory of Nonparametric Regression*. Springer, New York, 2002.

J. Klusowski and P. Tian. Large scale prediction with decision trees. *Journal of the American Statistical Association*, 119(545):525–537, 2024.

Jason Klusowski. Sharp analysis of a simple model for random forests. In *International Conference on Artificial Intelligence and Statistics*, pages 757–765. PMLR, 2021.

# References III



Balaji Lakshminarayanan, Daniel M Roy, and Yee Whye Teh. Mondrian forests: Efficient online random forests. *Advances in neural information processing systems*, 27, 2014.

X. Li, Y. Wang, S. Basu, K. Kumbier, and B. Yu. A debiased mdi feature importance measure for random forests. *NeurIPS*, 2019.

Yi Lin and Yongho Jeon. Random forests and adaptive nearest neighbors. *Journal of the American Statistical Association*, 101(474):578–590, 2006.

G. Louppe, L. Wehenkel, A. Sutera, and P. Geurts. Understanding variable importances in forests of randomized trees. In C.J.C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K.Q. Weinberger, editors, *Advances in Neural Information Processing Systems*, pages 431–439, 2013.

L. Mentch and G. Hooker. Quantifying uncertainty in random forests via confidence intervals and hypothesis tests. *JMLR*, 2016.

Jaouad Mourtada, Stéphane Gaïffas, and Erwan Scornet. Minimax optimal rates for mondrian trees and forests. *Annals of Statistics*, 2020.

E. Scornet. On the asymptotics of random forests. *Journal of Multivariate Analysis*, 146:72–83, 2016a.

# References IV



E. Scornet. Random forests and kernel methods. *IEEE Transactions on Information Theory*, 62(3):1485–1500, 2016b.

E. Scornet. Trees, forests, and impurity-based variable importance in regression. In *Annales de l'Institut Henri Poincare (B) Probabilites et statistiques*. Institut Henri Poincaré, 2023.

E. Scornet, G. Biau, and J.-P. Vert. Consistency of random forests. *The Annals of Statistics*, 2015.

Erwan Scornet and Giles Hooker. Theory of random forests. *Annual Review of Statistics and Its Application*, 13(1):99–121, 2026.

Yan Shuo Tan, Abhineet Agarwal, and Bin Yu. A cautionary tale on fitting decision trees to data from additive models: generalization lower bounds. In *International Conference on Artificial Intelligence and Statistics*, pages 9663–9685. PMLR, 2022.

S. Wager and S. Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 2018.

S. Wager and G. Walther. Adaptive concentration of regression trees, with application to random forests. *arXiv preprint arXiv:1503.06388*, 2015.

# Differences between uniform and non-uniform cases



Considering the centered forest with uniform covariates, we have

$$p_{k,\Theta}(\mathbf{x}) = \mathbb{P}(X \in L_\Theta(\mathbf{x})|\Theta) = 2^{-k}.$$

However, when the distribution is not uniform, we prove that

$$\begin{aligned} p'_{k,\Theta}(\mathbf{x}) &= \mathbb{P}(X' \in L_\Theta(\mathbf{x})|\Theta) \\ &= \frac{c'(\mathbf{x})}{2^k}\left(1 + \alpha'(\mathbf{x})\varepsilon'_\Theta(\mathbf{x})\mathrm{Diam}(L_\Theta(\mathbf{x}))\right) \qquad \text{a.s.} \end{aligned}$$

- Random variable and not a deterministic quantity
- Depends on $\mathbf{x}$

# Asymptotic constant



In our theoretical result for uniform covariates

$$\sqrt{\frac{nk^{(d-1)/2}}{2^k}}\left(\widehat{\mu}_s^{\mathrm{ICRF}}(\mathbf{x})-\mu(\mathbf{x})\right)\xrightarrow[n\to\infty]{d}\mathcal{N}(0,C(d)\mu(\mathbf{x})(1-\mu(\mathbf{x})))\,.$$

with

$$C(d)=\frac{2\Gamma(d-1)}{(\log 2)^{d-1}\Gamma((d-1)/2)}\mathbb{E}\left[\left(\frac{\|(\mathbf{N}-\overline{\mathbf{N}}\mathbb{1})\|_2}{\|(\mathbf{N}-\overline{\mathbf{N}}\mathbb{1})\|_1}\right)^{d-1}\right],$$

where $\mathbf{N}=(N_1,\ldots,N_d)$ with $N_1,\ldots,N_d$ independent $\mathcal{N}(0,1)$ and $\overline{\mathbf{N}}=(1/d)\sum_{j=1}^d N_j$.

This comes from a new control of the quantity: when $k\to\infty$,

$$\mathbb{E}\left[\mathbb{P}(X_1\in L_\Theta(\mathbf{x})|X_1)^2\right]\sim\frac{C(d)}{2^k k^{(d-1)/2}}.$$